Practical No.: 05

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COURSE: AI&DS
CLASS: BE
SUB: Computer Laboratory-IV
Title: Deep Convolutional Generative Adversarial Network (DCGAN) for Image Generation.

0.1 Import Required Libraries

[20]:
```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Reshape, Flatten
from tensorflow.keras.layers import Conv2D, Conv2DTranspose
from tensorflow.keras.layers import LeakyReLU, Dropout
from tensorflow.keras.optimizers import Adam

import numpy as np
import matplotlib.pyplot as plt
```

0.2 Load and Preprocess Dataset

[21]:
```
(x_train, _), (_, _) = mnist.load_data()

# Normalize images to [-1, 1]
x_train = (x_train.astype('float32') - 127.5) / 127.5

# Reshape to (28, 28, 1)
x_train = x_train.reshape(-1, 28, 28, 1)

print("Training data shape:", x_train.shape)
```

```
Training data shape: (60000, 28, 28, 1)
```

0.3 Define Generator Model

[22]:
```
def build_generator():
    model = Sequential()

    model.add(Dense(7*7*128, input_dim=100))
    model.add(LeakyReLU(0.2))
    model.add(Reshape((7, 7, 128)))

    model.add(Conv2DTranspose(64, kernel_size=4, strides=2, padding='same'))
```

```
    model.add(LeakyReLU(0.2))

    model.add(Conv2DTranspose(1, kernel_size=4, strides=2, padding='same',␣
 ↪activation='tanh'))

    return model

generator = build_generator()
generator.summary()
```

Model: "sequential_6"

Layer (type)	Output Shap	Param #
dense_4 (Dense)	(None, 627	633,472
leaky_re_lu_8 (LeakyReLU)	(None, 627	0
reshape_2 (Reshape)	(None, 7, 7 128)	0
conv2d_transpose_4 (Conv2DTranspose)	(None, 14, 14, 64)	131,136
leaky_re_lu_9 (LeakyReLU)	(None, 14, 14, 64)	0
conv2d_transpose_5 (Conv2DTranspose)	(None, 28, 28, 1)	1,025

Total params: 765,633 (2.92 MB)

Trainable params: 765,633 (2.92 MB)

Non-trainable params: 0 (0.00 B)

0.4 Define Discriminator Model

[23]:
```
def build_discriminator():
    model = Sequential()

    model.add(Conv2D(64, kernel_size=4, strides=2, padding='same',
                     input_shape=(28,28,1)))
    model.add(LeakyReLU(0.2))
```

```
    model.add(Dropout(0.3))

    model.add(Conv2D(128, kernel_size=4, strides=2, padding='same'))
    model.add(LeakyReLU(0.2))
    model.add(Dropout(0.3))

    model.add(Flatten())
    model.add(Dense(1, activation='sigmoid'))

    return model

discriminator = build_discriminator()
discriminator.summary()
```

Model: "sequential_7"

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 14, 14, 64)	1,088
leaky_re_lu_10 (LeakyReLU)	(None, 14, 14, 64)	0
dropout_4 (Dropout)	(None, 14, 14, 64)	0
conv2d_5 (Conv2D)	(None, 7, 7, 128)	131,200
leaky_re_lu_11 (LeakyReLU)	(None, 7, 7, 128)	0
dropout_5 (Dropout)	(None, 7, 7, 128)	0
flatten_2 (Flatten)	(None, 6272)	0
dense_5 (Dense)	(None, 1)	6,273

Total params: 138,561 (541.25 KB)

Trainable params: 138,561 (541.25 KB)

Non-trainable params: 0 (0.00 B)

0.5 Compile Discriminator

[24]:
```
discriminator.compile(
    loss='binary_crossentropy',
    optimizer=Adam(0.0002, 0.5),
    metrics=['accuracy']
)
```

0.6 Build and Compile GAN

[25]:
```
discriminator.trainable = False

gan = Sequential()
gan.add(generator)
gan.add(discriminator)

gan.compile(
    loss='binary_crossentropy',
    optimizer=Adam(0.0002, 0.5)
)
```

0.7 Training Function

[26]:
```
def train_dcgan(epochs=3000, batch_size=64, sample_interval=500):
    real = np.ones((batch_size, 1))
    fake = np.zeros((batch_size, 1))

    for epoch in range(epochs):
        # Train Discriminator
        idx = np.random.randint(0, x_train.shape[0], batch_size)
        real_imgs = x_train[idx]

        noise = np.random.normal(0, 1, (batch_size, 100))
        fake_imgs = generator.predict(noise, verbose=0)

        d_loss_real = discriminator.train_on_batch(real_imgs, real)
        d_loss_fake = discriminator.train_on_batch(fake_imgs, fake)

        # Train Generator
        noise = np.random.normal(0, 1, (batch_size, 100))
        g_loss = gan.train_on_batch(noise, real)

        if epoch % sample_interval == 0:
            print(f"Epoch {epoch} | D Loss: {d_loss_real[0]:.4f} | G Loss: {g_loss:.4f}")
            sample_images(epoch)
```

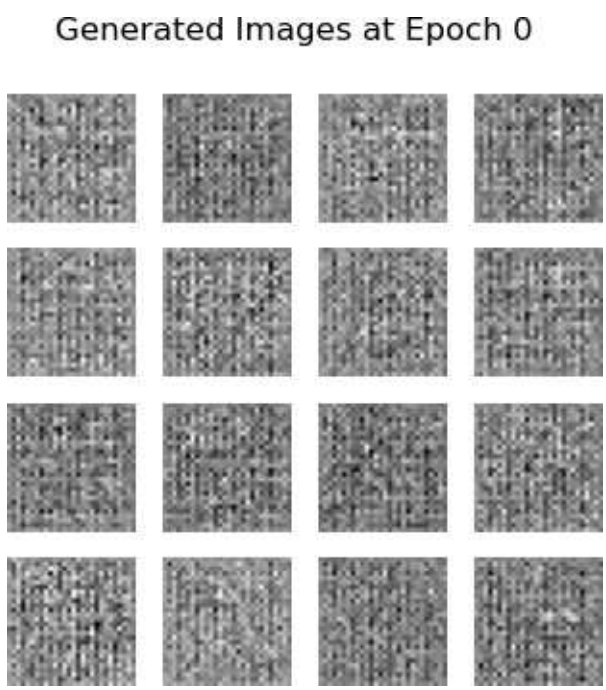
0.8 Image Sampling Function

[27]:
```
def sample_images(epoch):
    noise = np.random.normal(0, 1, (16, 100))
    gen_imgs = generator.predict(noise, verbose=0)

    gen_imgs = 0.5 * gen_imgs + 0.5   # Rescale to [0,1]

    fig, axs = plt.subplots(4, 4, figsize=(4,4))
    count = 0
    for i in range(4):
        for j in range(4):
            axs[i,j].imshow(gen_imgs[count, :, :, 0], cmap='gray')
            axs[i,j].axis('off')
            count += 1
    plt.suptitle(f"Generated Images at Epoch {epoch}")
    plt.show()
```

0.9 Train the DCGAN

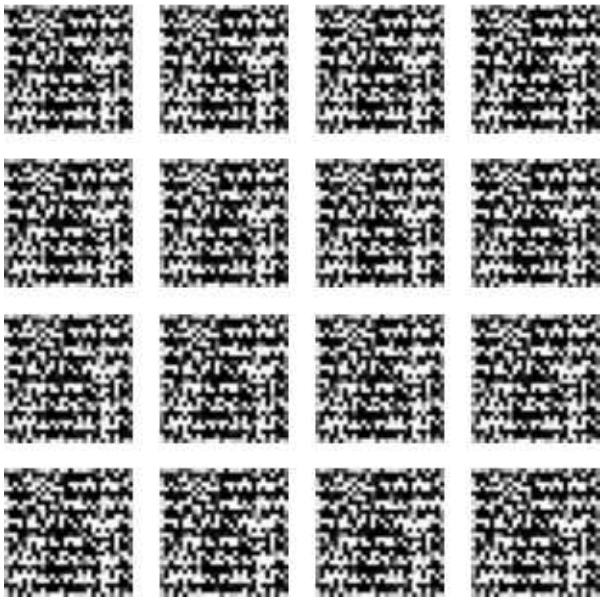
[28]:
```
train_dcgan(epochs=3000, batch_size=64, sample_interval=500)
```

Epoch 0 | D Loss: 0.6750 | G Loss: 0.6916



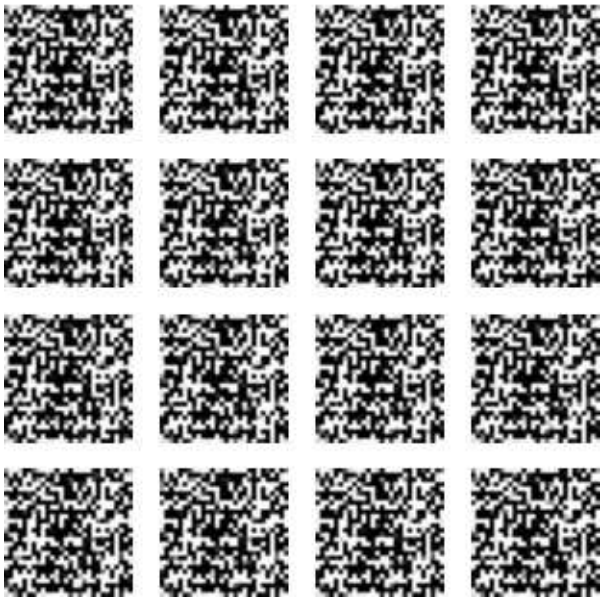
Epoch 500 | D Loss: 0.9356 | G Loss: 0.3846

Generated Images at Epoch 500



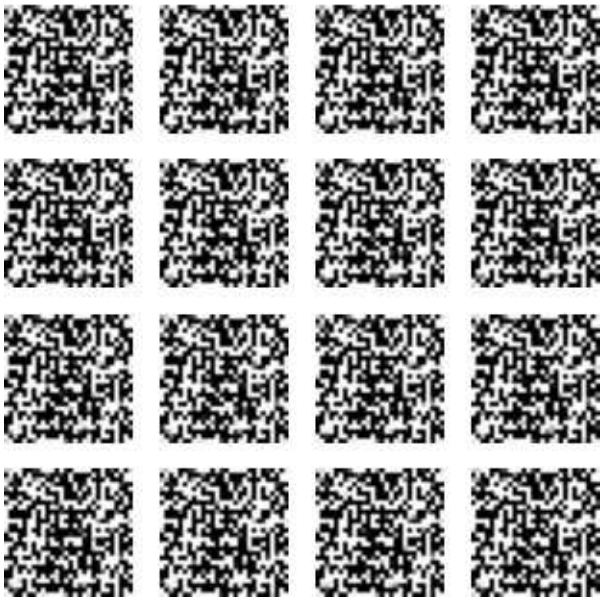
Epoch 1000 | D Loss: 1.0631 | G Loss: 0.2929

Generated Images at Epoch 1000



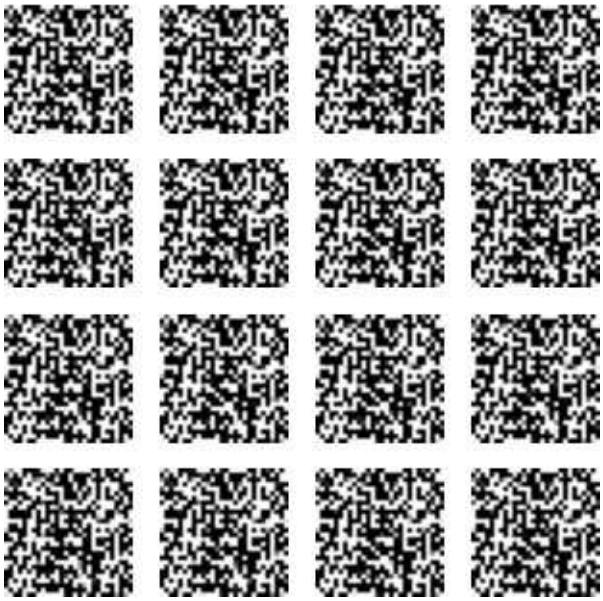
Epoch 1500 | D Loss: 1.1249 | G Loss: 0.2543

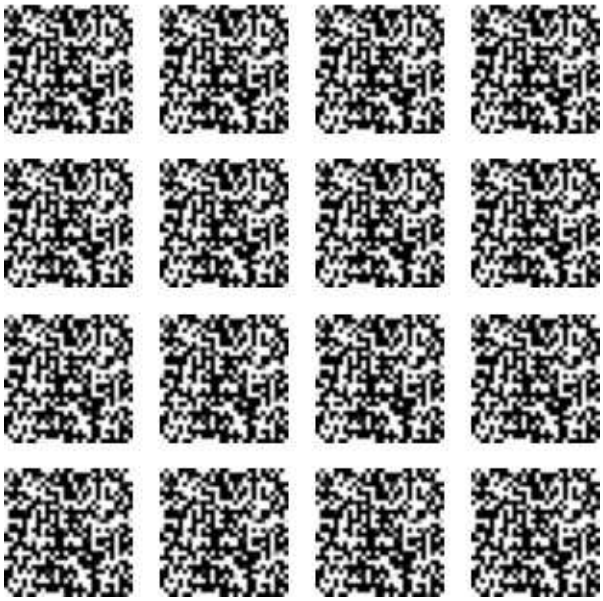
Generated Images at Epoch 1500



Epoch 2000 | D Loss: 1.1605 | G Loss: 0.2331

Generated Images at Epoch 2000



Epoch 2500 | D Loss: 1.1841 | G Loss: 0.2197

Generated Images at Epoch 2500